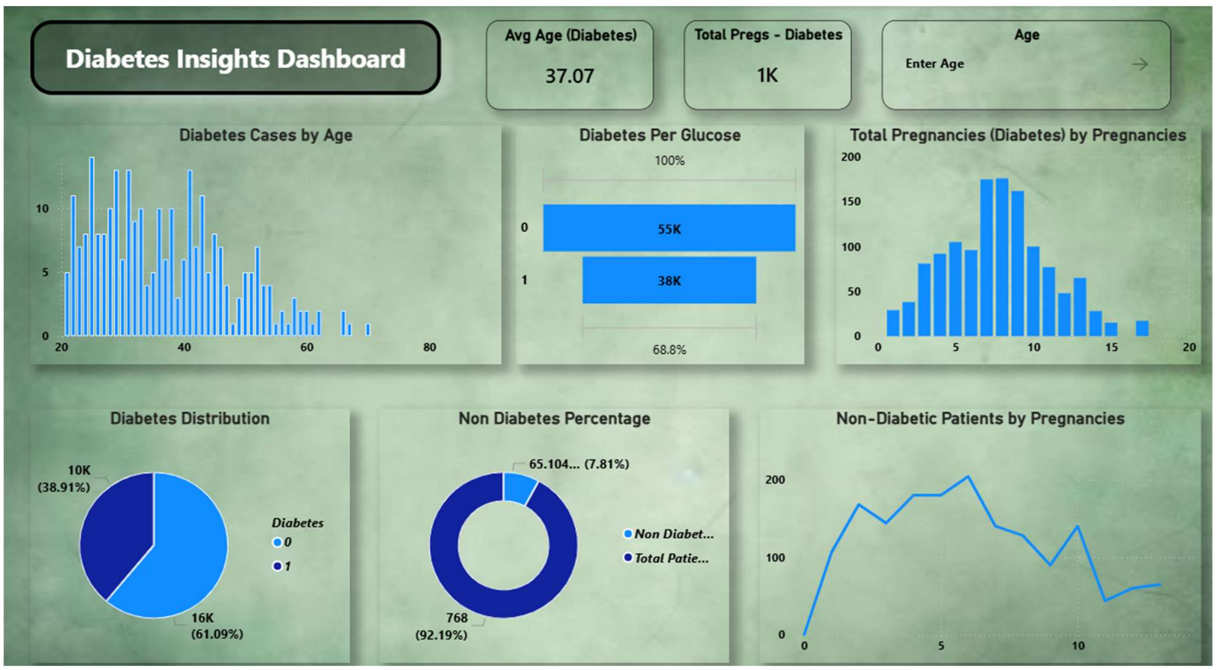
Report

A report is a comprehensive document that provides a detailed and structured account of data analysis, findings, and insights. It is typically used for in-depth analysis, documentation, and communication of results. Reports are suitable for a diverse audience, including decision-makers, analysts, and stakeholders who need a comprehensive understanding of the data.

Designing a report in Power BI involves connecting to data sources, creating visualizations like charts and graphs, customizing their appearance and interactivity, organizing them logically on the canvas, formatting elements for consistency and clarity, and optionally creating dashboards for a summarized view. Throughout the process, it's essential to consider the audience's needs and ensure the report effectively communicates insights from the data. Finally, iterate based on feedback to continually improve the report's design and usefulness.



Observations drawn from reports in Power BI can provide valuable insights into malnutrition trends:

Diabetes Insights Dashboard – Executive Report

Overview

This dashboard analyzes patient health records to identify patterns associated with diabetes prevalence. Key variables include age, glucose level, and pregnancy history.

Metric	Value	Insight
Total Patients	768	Dataset Size
Diabetic Patients	268 (38.91%)	Significant diabetes burden
Non-Diabetic Patients	500 (61.09%)	Majority of patients
Average Age (Diabetic)	37.07 Years	Diabetes is common among middle-aged adults
High-Risk Group	Age 30–50	Increased prevalence observed

Figure 1: Diabetes Cases by Age
Observation
The majority of diabetes cases are concentrated between **25 and 50 years of age**.
Interpretation
- Diabetes prevalence rises steadily during adulthood.
- Peak occurrences are observed among middle-aged individuals.
- Cases decline after age 60 due to smaller population representation.

Business/Healthcare Insight
Healthcare providers should prioritize diabetes screening programs for adults aged 30–50 years.

Figure 2: Diabetes per Glucose Level
Observation
Patients with elevated glucose levels contribute to the majority of diabetes diagnoses.
Interpretation
- Higher glucose concentration directly correlates with diabetes occurrence.
- Glucose appears to be the strongest predictive feature in the dataset.

Healthcare Insight
Routine blood glucose monitoring can significantly improve early diabetes detection.

Figure 3: Diabetes by Pregnancies
Observation
Diabetes cases are highly concentrated between **6 and 10 pregnancies**.
Interpretation
- Pregnancy history may influence insulin resistance.
- Higher pregnancy counts show increased diabetes prevalence.

Healthcare Insight
Women with multiple pregnancies should undergo regular diabetes screening.

Figure 4: Diabetes Distribution (Pie Chart)
Observation
- Diabetic Patients: 38.91%
- Non-Diabetic Patients: 61.09%

Interpretation
Nearly 4 out of every 10 patients in the dataset are diabetic.
Healthcare Insight
The dataset reflects a substantial diabetes burden, highlighting the importance of preventive healthcare measures.

Figure 5: Non-Diabetes Percentage
Observation
Approximately 61.09% of patients are classified as non-diabetic.
Interpretation
Most patients currently show no signs of diabetes, but preventive monitoring remains essential.
Recommendation
Implement awareness campaigns focused on maintaining healthy glucose levels and body weight.

Figure 6: Non-Diabetic Patients by Pregnancies
Observation
Non-diabetic patient counts remain relatively high for pregnancy counts between 3 and 8.
Interpretation
Although pregnancy count may contribute to diabetes risk, it alone does not determine diabetic status.
Recommendation
Combine pregnancy history with glucose and age data for more accurate risk assessment.

High-Risk Patient Profile
Based on the dashboard analysis, a typical high-risk patient exhibits:
Demographic Characteristics
- Age: 30–50 Years
- Female with multiple pregnancies
- Elevated glucose levels

Clinical Indicators
- High blood glucose concentration
- Pregnancy count above 5
- Family history (if available)

Risk Level
High
Suggested Action
Immediate diabetes screening and lifestyle intervention.

Final Conclusion
Overall Findings
1. **38.91% of patients are diabetic**, indicating a considerable disease prevalence.
2. **Average diabetic age is 37.07 years**, suggesting diabetes affects working-age adults.
3. **Glucose level is the strongest risk factor** associated with diabetes.
4. **Pregnancy history shows a positive correlation** with diabetes occurrence.
5. **Middle-aged individuals with high glucose levels and multiple pregnancies represent the highest-risk group.**

Strategic Recommendation
Healthcare organizations should focus on:
- Early glucose screening programs.
- Targeted interventions for adults aged 30–50 years.
- Regular monitoring of women with multiple pregnancies.
- Awareness campaigns promoting healthy lifestyles and diabetes prevention.